MEMORANDUM

To: Vice President, Population Health Group, Meridian Care Network
From: Data Analytics Team
Subject: 30-Day Diabetes Readmission Analysis: Findings and Recommendations
Date: 31 October 2026

1. Executive Summary

This analysis examines factors associated with 30-day hospital readmissions among patients with diabetes, with the objective of identifying encounters that may warrant prioritised post-discharge care coordination.

Following the exclusion of 2,423 encounters based on discharge disposition, the eligible analytical cohort comprised **99,343 encounters**, with 11,314 readmissions within 30 days, representing an overall readmission rate of **11.39%**.

The analysis identified several characteristics associated with elevated readmission rates, particularly previous inpatient admissions, emergency department utilisation, higher medication counts, greater diagnostic complexity and certain indicators of clinical severity.

An explainable, points-based risk-stratification model was subsequently developed to identify encounters with comparatively elevated readmission rates and support more targeted allocation of care-coordination resources.

2. Key Findings

A. Patient demographics and admission characteristics

- Young adults aged 20–30 recorded the highest observed readmission rate among the age groups examined. Older age groups showed relatively stable rates, generally close to the overall cohort average.
- Emergency admissions recorded the highest readmission rate among the major admission categories at **11.83%**, compared with 11.35% for urgent admissions and 10.49% for elective admissions.
- Emergency admissions also accounted for 6,194 of the 11,314 recorded 30-day readmissions, making them the largest contributor to the readmission count.
- Rare categories, including newborn and trauma centre admissions, and unmapped classifications were treated cautiously because of small sample sizes or classification limitations.

B. Clinical indicators and healthcare utilisation

Several variables demonstrated meaningful associations with 30-day readmission:

- **Previous inpatient admissions:** Readmission rates increased substantially with previous inpatient utilisation. Encounters with no previous inpatient visits recorded an 8.59% rate, compared with 24.14% for four previous visits. Rates reached 31.98%–46.21% among encounters with five to eight previous visits.
- **Emergency department utilisation:** Encounters with no previous emergency visits recorded a 10.69% readmission rate. This increased to 14.68% for one previous visit and 18.75% for two. The rate reached 30.89% among encounters with four previous visits, although this category contained only 369 encounters.
- **Medication burden:** Higher medication counts were generally associated with higher readmission rates. The observed rate increased from 7.54% at five medications to 14.14% at 24 medications and 17.30% at 35 medications. The relationship was not perfectly linear, particularly at higher counts.
- **Diagnostic complexity:** Readmission rates generally increased with the number of diagnoses, rising from 5.94% among encounters with one diagnosis to 12.76% among those with nine.
- **Length of stay:** Encounters with stays of 7–9 days recorded a 13.50% readmission rate, compared with 9.69% for stays of 1–3 days. The rate declined slightly to 12.76% for stays of 10 days or more.

Additional clinical observations included a 15.07% readmission rate among encounters in the maximum recorded glucose category (>300), compared with 12.97% for >200, 11.55% for normal, and 11.31% for none. Encounters without a recorded HbA1c result had the highest observed rate among the HbA1c recording categories, at 11.68%. These findings warrant further investigation but do not establish causation.

Previous outpatient visits and procedure counts did not show similarly consistent increases in readmission.

3. Risk Stratification and Resource Allocation

Based on the exploratory findings, an explainable risk score was developed using five encounter characteristics:

- Previous inpatient admissions.
- Previous emergency visits.
- Length of hospital stay.
- Number of diagnoses.
- Number of medications.

Each characteristic contributes points according to predefined thresholds, producing a cumulative score from 0 to 9.

Encounters were classified into three tiers:

Risk tier	Score	Encounters	30-day readmission rate
Low Risk	0–3	49,492	8.26%
Moderate Risk	4	20,230	11.52%
High Risk	5–9	29,621	16.54%

The High Risk tier represents approximately **29.82% of eligible encounters**, yet accounts for **43.29% of all observed 30-day readmissions** (4,898 of 11,314).

This indicates that the model concentrates a disproportionate share of observed readmissions within a smaller group of encounters, making it a potentially useful mechanism for prioritising care-coordination attention.

4. Estimated Resource Implications

To illustrate the potential operational implications, the analysis assumes one hour of care coordination per eligible encounter.

Under this assumption:

- Coordinating all 99,343 eligible encounters would require an estimated 99,343 hours.
- Coordinating only the 29,621 High Risk encounters would require an estimated 29,621 hours.
- The difference is 69,722 hours, representing a **70.18% reduction in estimated coordination workload** relative to coordinating every eligible encounter.

However, this approach has an important trade-off: 56.71% of observed 30-day readmissions occurred outside the High Risk tier.

The proposed prioritisation strategy should therefore not be interpreted as identifying every encounter that will result in readmission, nor as evidence that the remaining readmissions are preventable. The workload estimate is illustrative and does not represent measured staffing savings or demonstrated reductions in readmission.

5. Recommendations

Based on the findings, the following actions are recommended:

1. Pilot risk-based care coordination.
Meridian should consider piloting the proposed High Risk classification as a prioritisation tool for post-discharge care coordination, subject to clinical review and validation.

2. Prioritise encounters with repeated hospital utilisation.
Previous inpatient and emergency department visits demonstrated particularly strong associations with subsequent readmission. These indicators warrant attention when assessing patients' post-discharge support needs.

3. Investigate clinical complexity and treatment burden.
Higher medication counts, greater numbers of diagnoses and longer hospital stays were associated with elevated readmission rates. Further investigation could help identify the underlying clinical and care-management needs associated with these patterns.

4. Maintain a safety net for encounters outside the High Risk tier.
Because the High Risk group contains 43.29% of observed readmissions, an exclusive focus on this group would leave a substantial proportion of readmissions outside the prioritised population. Meridian should consider a secondary screening or escalation pathway for Moderate and Low Risk encounters.

5. Validate the model before operational deployment.
The risk score was developed using historical observational data from 1999–2008. It should be tested against more recent patient data and evaluated for predictive performance, calibration, subgroup differences and operational feasibility before being adopted as a clinical decision-support tool.

6. Conclusion

The analysis suggests that a targeted approach to care coordination may offer Meridian an opportunity to concentrate limited resources on encounters with higher observed readmission rates.

The proposed risk-stratification model identifies a group representing 29.82% of eligible encounters but containing 43.29% of observed 30-day readmissions. Under the stated workload assumption, prioritising this group could reduce the estimated coordination workload by 70.18%.

The recommended next step is a carefully monitored pilot, supported by clinical oversight, further validation, and a safety net for patients outside the High Risk tier.